VIVEK SINGH

AUTOMATION ENGINEER

CONTACT

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Portfolio :- https://portfolio-tawny-alpha-96.vercel.app/

EXPERTISE

- Management Skills, UML
- Python, Django, FASTAPI, TensorFlow, RESTFUL API, Keras, PyTorch
- Transformer's Architecture, U-NET Architecture, Deep Learning, Machine Learning, Agentic AI, GENAI
- ReactJS, UI/UX, Full Stack Development
- Vector DB, Oracle DB, Docker
- AI Algorithm's , Data Science, NLP, LLM, RAG, DSA(Data Structure & Algorithm's).

LANGUAGE

- English
- Hindi

WORK EXPERIENCE

Automation Engineer
Bharti Airtel - Gurugram
2022 - Present

Principal AI Engineer and Software Development Leader specializing in the architecture and implementation of future-ready, intelligent systems. Deep expertise in full-stack development, cloud computing, and advanced Machine Learning, with a focus on integrating Generative AI, Agentic AI, and proprietary AI Algorithms for actionable insights and product management. Proven ability in secure system integrations (LDAP, SSO), Power BI analytics, database management, and automating system operations using Selenium. Committed to solving complex business problems through rigorous architectural design and optimized data flows.

EDUCATION

Galgotia's College Of Engineering and Technology

Bachelor of Technology (B.tech)
2018-2022

PROJECTS

Ai Insight: - The project is a custom AI-powered chatbot designed specifically for network engineers and operations teams. It integrates AirLM (LLM) with Retrieval-Augmented Generation (RAG) to provide intelligent, context-aware responses. The chatbot assists with troubleshooting, network optimization, and insights by retrieving relevant information from a Vector Database (Chroma DB) and combining it with the reasoning capabilities of AirLM. A reinforcement learning loop is incorporated by capturing user feedback, which continuously fine-tunes AirLM to improve accuracy, personalization, and relevance over time. The system is deployed through FastAPI as the backend inference engine and ReactJS as the frontend interface, ensuring scalability, performance, and user-friendly interactions. This end-to-end solution not only empowers network engineers with real-time insights and automated recommendations but also evolves adaptively based on user feedback, ensuring long-term efficiency and reliability.

AirLM: - This project involves the development of a custom Large Language Model (LLM) built on Transformer architecture from the ground up. The model is designed to process natural language efficiently while incorporating reasoning capabilities and reinforcement learning (RL) for adaptive improvement. Unlike generic pre-trained models, this LLM is domain-specific and fully custom-built, ensuring fine-grained control over architecture, training, and inference. The implementation leverages TensorFlow and Keras for neural network development, FastAPI for serving the model as an inference API, and ReactJS for building an interactive user interface. The model's workflow includes tokenization, embeddings, positional encoding, attention mechanisms, and stacked transformer layers, leading to a decoder-based inference system. Additional files such as config. json and vocab. json are generated to facilitate portability, reproducibility, and ease of deployment. By adding reasoning modules and reinforcement learning (RLHF-inspired adaptive learning), the system evolves dynamically, fine- tuning outputs based on user interactions and feedback. This results in a self-improving AI assistant capable of delivering context- aware, adaptive, and optimized responses.

PixelEngine: - The core of this project is to repurpose or extend a trained U-Net image segmentation model for new generative tasks. a. Core Model: A U-Net architecture, typically trained on an image segmentation task (e.g., segmenting objects in medical images, or background/foreground). b. Adaptation/Extension: Network Topology Generation: The U-Net's segmentation capability (identifying distinct regions/classes) can be adapted to represent and generate network components (routers, switches, links) as distinct segments on a 'canvas' image. Live Image Generation on Prompt: This is a more complex task. The U-Net could be used as the decoder part of a larger Generative Adversarial Network (GAN) or Diffusion Model. The prompt (text) would be encoded into a latent vector, which then guides the U-Net/decoder to construct the final image.

NetVision: - Developed a real-time network monitoring and analytics platform that ingests syslog data from network devices using Apache Kafka, processes it through an ETL pipeline, and stores structured data in a database for advanced analytics and visualization. The system categorizes events based on Severity, Alarm Types, and Event Patterns to provide operational visibility.
Implemented an AI-powered insights engine using an MCP server and the AirLM model to automatically generate contextual insights from the processed data. The system uses adaptive prompt engineering, where prompts are dynamically generated based on the incoming dataset instead of user-provided queries, enabling automated anomaly interpretation and operational recommendations.
Built APIs to deliver insights and integrated them with an interactive data visualization dashboard for real-time monitoring of alarms, severity trends, and network events.

Unified Data Inventory:- Developed a comprehensive ticketing analytics platform that provides a centralized, aggregated view of Remedy tickets across four key domains: Homes, Infra, Service, and Passive. The system enables stakeholders to monitor operational health by visualizing ticket distributions across Site Groups, Assigned Groups, and total volumes. To enhance user efficiency, I integrated a Natural Language AI Search feature, allowing users to instantly retrieve granular ticket data through simple queries or ticket IDs. Furthermore, I implemented an AI Insights engine powered by MCP (Model Context Protocol) Servers, utilizing specialized AI Agents to perform deep-dive analysis and trend reporting on cross-domain data.

MOP.ai :- Developed an AI-based application for automated generation of Methods of Procedure (MOP).
Created a comprehensive repository of standardized MOP documents for training and reference.
Designed and trained an AI model using a transformer-based architecture.
Leveraged the repository to enable accurate learning of MOP structure and content.
Integrated the trained model into an application pipeline.

Nexus: -Developed a full-stack, high-performance dashboard using ReactJS and an asynchronous FASTAPI backend with an integrated inference engine. The system fetched real-time operational data from Remedy tickets for BNG & QFX devices. It provided dynamic visualizations of key metrics including Aging, SLA compliance, and categorical data. A core feature was the implementation of a Long Short-Term Memory (LSTM) AI model for real-time prediction of future ticket aging and SLA breach volume. This solution delivered proactive insights, significantly enhancing operational efficiency and service management.

NOC Beats: -The project is an Interface and Data Lake solution for managing and analyzing CDR (Call Detail Records). It focuses on monitoring MSAN (Multi-Service Access Node) details, DSL IDs, and BNG (Broadband Network Gateway) Interfaces. The system captures Record Types and Record Dates, storing dropped BNG/MSAN details in a centralized Data Lake for analysis, troubleshooting, And decisions making. The solution enables: Real-time collection and storage of CDR data Integration of MSAN and BNG network elements for performance monitoring A web-based interface for visualization and insights APIs for data retrieval and system interoperability.

Airtel Image Manager (AIM) : - Designed and implemented a custom interface to replace the Cisco SWIM interface, enabling efficient and automated upgrades across multiple vendor routers, including Cisco, Nokia, Huawei, Juniper, and Ciena.